

SKILLZENO INTERNSHIP

TASK 2: BATTERY ANALYSIS

Electric Vehicle Technology Internship

Name	K. Reddi Rubia
Roll No.	23KA1A0225
College	JNTUA College of Engineering, Kalikiri
Branch	Electrical and Electronics Engineering (EEE)
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Technical Report

Lithium-ion EV battery pack study, sizing calculations, configuration comparison and BMS analysis

1. Introduction

Electric vehicles (EVs) depend on rechargeable battery packs as their primary energy source. Lithium-ion batteries are widely used because they offer high energy density, good efficiency, relatively low self-discharge, and long cycle life when operated within safe limits. An EV battery pack is built from many individual cells connected in series and parallel so that the required pack voltage, capacity and energy can be achieved.

1.1 Major Parts of a Lithium-ion EV Battery Pack

- Cells: the basic electrochemical units that store electrical energy.
- Modules: groups of cells mechanically and electrically assembled together.
- Battery pack: a collection of modules, busbars, fuses, contactors, sensors and enclosure.
- Thermal management system: maintains a safe operating temperature using air or liquid cooling.
- Battery Management System (BMS): monitors, protects and balances the cells and coordinates charging/discharging.

1.2 Common Lithium-ion Chemistries in EVs

Chemistry	Typical Strength	Main Limitation	EV Use
NMC	High energy density and power	Cost and thermal sensitivity	Many passenger EVs
NCA	Very high specific energy	Requires careful thermal control	Long-range EV applications
LFP	Long cycle life and good thermal stability	Lower energy density	Buses, entry EVs, stationary storage

2. Battery Capacity and Energy Requirements

Battery sizing starts with the energy required by the vehicle. The most common quantities are pack voltage (V), capacity in ampere-hours (Ah), and energy in watt-hours (Wh) or kilowatt-hours (kWh).

Energy	$E \text{ (Wh)} = V \text{ (V)} \times \text{Capacity (Ah)}$
Capacity	$\text{Capacity (Ah)} = E \text{ (Wh)} / V \text{ (V)}$
Driving energy	$\text{Required energy (kWh)} = \text{Range (km)} \times \text{Consumption (kWh/km)}$

2.1 Worked Example

Assume an EV requires 30 kWh of usable battery energy and uses a nominal 300 V battery pack.

- Convert energy to watt-hours: $30 \text{ kWh} = 30,000 \text{ Wh}$.
- Capacity = $30,000 \text{ Wh} / 300 \text{ V} = 100 \text{ Ah}$.
- Therefore, a 300 V, 100 Ah battery pack stores approximately 30 kWh of nominal energy.

If a 10% energy reserve is desired so the pack is not normally discharged to 0%, the nominal pack energy should be about $30 / 0.90 = 33.3 \text{ kWh}$. This reserve helps reduce deep discharge stress and provides operating margin.

3. Comparison of Two Battery Configurations

Two packs can store the same total energy even when their voltages and ampere-hour capacities are different. The following example compares two 30 kWh configurations.

Parameter	Configuration A	Configuration B	Observation	Result
Nominal voltage	300 V	400 V	B has higher voltage	-
Capacity	100 Ah	75 Ah	A has higher Ah	-
Stored energy	30 kWh	30 kWh	Same energy	Equal
Current at 60 kW	200 A	150 A	B needs less current	B lower
I^2R loss trend	Higher	Lower	Lower current reduces conductor loss	B advantage

3.1 Current Comparison at the Same Power

Electrical power is approximately $P = V \times I$. For a 60 kW demand:

- Configuration A: $I = 60,000 / 300 = 200$ A.
- Configuration B: $I = 60,000 / 400 = 150$ A.
- At the same power, the higher-voltage configuration carries 25% less current. Lower current can reduce resistive losses, heat generation and conductor size, although higher voltage requires suitable insulation, contactors and safety design.

3.2 Series and Parallel Connections

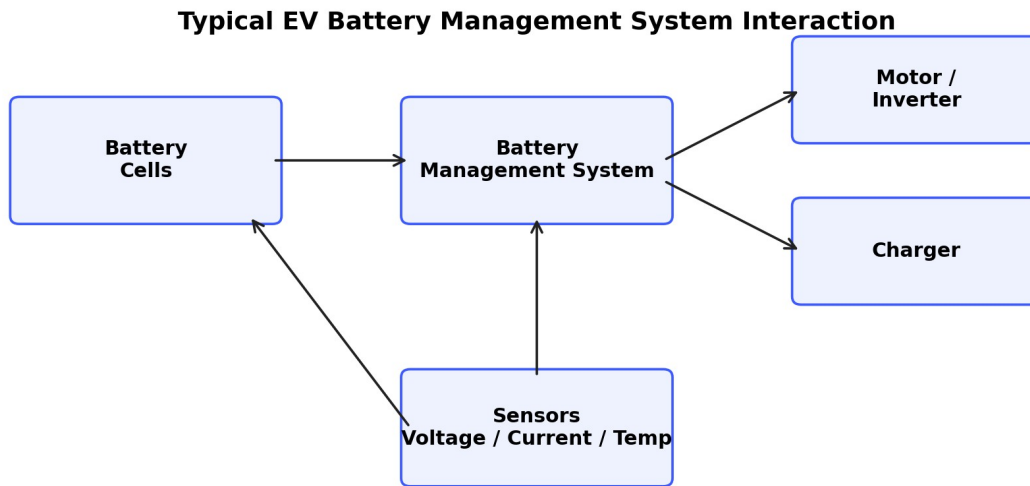
Cells connected in series increase pack voltage, while cells connected in parallel increase ampere-hour capacity. For example, if a nominal 3.7 V, 5 Ah cell is used, the approximate cell counts are:

- For 300 V: about 81 cells in series ($300 / 3.7 \approx 81$).
- For 400 V: about 108 cells in series ($400 / 3.7 \approx 108$).
- To reach 100 Ah using 5 Ah cells: 20 cells in parallel.
- To reach 75 Ah using 5 Ah cells: 15 cells in parallel.

These are simplified nominal calculations. Real EV packs use manufacturer-defined voltage windows, module layouts, safety margins and exact series/parallel counts.

4. Battery Management System (BMS)

The BMS is the electronic control system responsible for keeping the EV battery within safe electrical and thermal limits. Because a lithium-ion pack contains many cells, small differences between cells can grow over time. The BMS continuously measures important parameters and takes protective action when required.



The BMS monitors cell conditions and controls charging/discharging to protect the battery pack.

Figure 1. Typical interaction of an EV Battery Management System.

4.1 Important BMS Functions

- Cell voltage monitoring: detects over-voltage and under-voltage conditions.
- Current monitoring: measures charge and discharge current and detects over-current or short-circuit conditions.
- Temperature monitoring: prevents operation outside safe thermal limits.
- State of Charge (SOC) estimation: estimates the remaining usable charge in the battery.
- State of Health (SOH) estimation: tracks long-term battery degradation and remaining performance.
- Cell balancing: keeps cell voltages closely matched so one weak or highly charged cell does not limit the whole pack.
- Contactor control: safely connects or disconnects the high-voltage battery from the vehicle.
- Communication: exchanges data with the vehicle control unit, inverter and charger.

4.2 Why the BMS is Essential

Lithium-ion cells can be damaged by overcharging, deep discharging, excessive current and high temperature. A correctly designed BMS improves safety, protects the pack, supports consistent vehicle performance, enables accurate range estimation and helps extend battery service life.

5. Summary and Conclusion

Lithium-ion battery packs are central to EV performance because they determine driving range, power capability, charging behaviour and vehicle mass. Battery energy is obtained from pack voltage and ampere-hour capacity, while required capacity can be estimated from the vehicle energy demand. In the worked example, a 300 V, 100 Ah pack provides approximately 30 kWh. A 400 V, 75 Ah configuration provides the same energy but requires lower current at the same power level.

The comparison shows why modern EVs often use higher-voltage architectures: lower current can reduce resistive losses and heat, although the system must be designed for the higher insulation and safety requirements. The BMS is essential for safe battery operation because it monitors voltage, current and temperature, estimates SOC/SOH, balances cells and disconnects the pack when unsafe conditions occur.

Learning Outcomes

- Understood the construction and role of lithium-ion battery packs in EVs.
- Calculated battery energy and ampere-hour capacity using basic electrical relations.
- Compared two battery configurations with equal energy but different voltage and capacity.
- Explained the major monitoring, protection and balancing functions of a BMS.

Key Formula Reference

Quantity	Formula	Unit
Energy	$E = V \times Ah$	Wh
Capacity	$Ah = Wh / V$	Ah
Current	$I = P / V$	A
Resistive loss	$P_{loss} = I^2 R$	W

Prepared by: K. Reddi Rubia (Roll No. 23KA1A0225)

Institution: JNTUA College of Engineering, Kalikiri